

Subtype Stratification of Parkinson’s Disease and Longitudinal Signatures across PPMI and OPDC

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This work presents an integrated pipeline for subtype stratification of Parkinson’s disease at baseline and the characterisation of longitudinal progression signatures across two large cohorts (PPMI and OPDC). The framework combines baseline clinical clustering, cross-cohort harmonisation, and longitudinal as well as biological validation to derive subtypes that are statistically stable, clinically interpretable, and supported by independent biomarkers.

The analysis pools 821 de novo Parkinson’s patients from PPMI and 146 early Parkinson’s patients from OPDC, selected under levodopa-naïve or near-naïve conditions with harmonised inclusion criteria. Nine shared baseline clinical markers spanning motor function (MDS-UPDRS I-III), cognition (MoCA), functional autonomy (Schwab-England), sleep and REM behaviour (RBDSQ, ESS), autonomic function (orthostatic blood pressure drop, SCOPA-AUT), and anxiety (HADS-A) are used as inputs for subtype derivation. A two-step pre-processing pipeline first applies GAMLSS-based normalisation on a prodromal reference population, conditioning on age, sex and education to remove demographic confounding and express all clinical measures on a common z -score scale. Multiple imputation by chained equations (MICE) is then performed on these adjusted z -scores, under a working MAR assumption, to handle low levels of missingness (2.4% in PPMI, 1.3% in OPDC) while operating in an already harmonised biomarker space.

Between-cohort harmonisation is achieved using ComBat-EB, with cohort/site treated as batch and the clinical z -scores as features, to remove residual site effects while preserving biological signal. Gaussian mixture models (GMMs) with tied covariance are then fitted at baseline on PPMI for $K = 2, \dots, 8$ clusters, and 84 candidate clustering configurations are explored. These configurations are aggregated via a consensus bootstrap procedure (500 resamples) to obtain a stable consensus partition, with internal stability quantified by a bootstrap-adjusted Rand index ARI_{boot} (1,000 replications). An ablation study shows that GAMLSS normalisation provides the largest improvement in within-cohort coherence (silhouette gain of +0.157 relative to raw data), ComBat-EB yields the largest gain in cross-cohort adjusted Rand index ($\Delta \text{ARI}_{\text{xc}} = +0.084$), and the consensus step further increases ARI_{xc} from 0.130 to 0.178 without degrading the silhouette.

A composite criterion is used to select the final number of clusters, balancing clinical interpretability, cross-cohort reproducibility (ARI_{xc}), silhouette, and coverage of recognised clinical archetypes. While $K = 3$ maximises ARI_{xc} (0.394), $K = 5$ is retained because it recapitulates and refines five well-described archetypes (including “mild motor-predominant” and “diffuse malignant”) and provides superior separation of Kaplan-Meier survival curves for multiple baseline-defined clinical milestones. The final baseline five-cluster solution, obtained with tied-covariance GMMs, MICE, and ComBat-EB, achieves an inter-cohort $\text{ARI}_{\text{xc}} = 0.234$ and strong internal bootstrap stability ($\text{ARI}_{\text{boot}} = 0.726$, 95% CI 0.430-0.930).

In PPMI, the five baseline subtypes form a clinically ordered severity spectrum that is qualitatively reproduced when the PPMI-derived clustering solution is projected onto OPDC, despite different recruitment profiles. One subtype corresponds to a severe “diffuse malignant” profile with the most extreme deviations across motor, cognitive and autonomic domains (very high MDS-UPDRS II-III, marked Schwab-England decline, pronounced MoCA impairment) and exhibits the fastest deterioration across all outcome dimensions. Another frequent subtype

corresponds to a mild motor-predominant phenotype with relatively low motor scores, preserved autonomy and limited non-motor burden, while intermediate subtypes differ primarily in the extent of functional decline and non-motor symptoms. Cross-cohort projection of OPDC into the PPMI-derived clinical space confirms that this hierarchy of severity is preserved, although OPDC is enriched in severe cases and PPMI in de novo, more moderate phenotypes.

Geometric and qualitative coherence of the baseline clustering is assessed using PCA projections and consensus co-association matrices. In the PPMI-based PCA space, the five subtypes occupy partially separated regions along a primary motor-functional gradient and a secondary non-motor gradient, with the most severe subtype located at the extreme of both axes. Consensus co-association matrices display dense diagonal blocks for three of the five subtypes in PPMI and a particularly robust block for the severe subtype in OPDC, consistent with the high ARI_{boot} and indicating that instability arises mainly in the least populated subtypes of the smaller OPDC cohort.

External convergent validity is examined at several levels using markers that were not used for clustering at baseline. In PPMI, a suite of orthogonal clinical measures (e.g. SDMT, SCOPA-AUT total and sub-scores, QUIP-ICD) shows strong between-subtype differentiation (Kruskal-Wallis p -values often below 0.001 after Bonferroni correction), with the severe subtype displaying the lowest cognitive processing speed and the highest burden of autonomic dysfunction and impulsive-compulsive behaviours. Striatal DaTSCAN uptake measured at baseline and longitudinally up to three years demonstrates a graded pattern of nigrostriatal denervation: the severe subtype shows the lowest putaminal binding, whereas the mild motor-predominant subtype maintains relatively preserved striatal uptake. Exploratory SWI- R_2 iron imaging at reference (V04) suggests early differentiation in left putaminal iron deposition between subtypes; these MRI results are treated as exploratory because multiple testing and limited sample sizes preclude firm conclusions.

Longitudinal clinical trajectories are analysed post hoc, conditional on the baseline clustering, both as continuous time courses and via Kaplan-Meier survival for pre-defined clinical milestones. Mean trajectories over up to 17 PPMI visits reveal that the severe baseline subtype experiences a rapid, approximately linear decline in motor function and autonomy, accompanied by pronounced cognitive deterioration, whereas the mild motor-predominant and function-preserved subtypes remain comparatively stable. In both PPMI and OPDC, the severe subtype consistently shows the steepest deterioration in motor, functional and anxiety measures, while milder subtypes display more favourable and roughly parallel trajectories, despite wider confidence intervals in OPDC due to smaller numbers. Kaplan-Meier analyses for five clinically relevant endpoints (severe motor impairment, cognitive impairment, daytime sleepiness, significant dysautonomia and REM sleep behaviour disorder) yield globally significant log-rank tests ($p < 0.0001$ for all milestones), indicating that the baseline subtypes differ not only at inclusion but also in their subsequent time-to-event risk profiles.

Progression sequences associated with these baseline-defined subtypes are further characterised using a z -score-based SuStaIn model applied to the same set of baseline-normalised clinical z -scores. A 10-fold cross-validated information criterion selects a four-subtype SuStaIn model as optimal, which is trained on PPMI and then transferred to OPDC without retraining to assess the generalisability of progression patterns. The SuStaIn sequences reveal two “body-first” progression patterns, in which motor impairment (UPDRS-III) is the earliest event, and two “brain-first” patterns, in which loss of functional autonomy and anxiety precede motor symptoms, consistent with brain-first/body-first hypotheses for α -synuclein spread. The most advanced SuStaIn subtype combines a high average stage with multidomain impairment, aligning with the clinically severe baseline cluster and supporting the biological coherence of the stratification.

Concordance between the GMM-derived baseline subtypes and the SuStaIn progression subtypes is examined via cross-tabulation. Early, function-preserved GMM subtypes map predominantly to early SuStaIn stages, whereas more impaired GMM subtypes distribute across later SuStaIn trajectories, indicating that the baseline taxonomy reflects distinct progression

regimes rather than purely cross-sectional differences. Orthogonal genetic validation, leveraging a polygenic risk score ($\text{PRS}_{\text{META5PGS}}$) computed for 804 genotyped PPMI participants, shows a significant difference in PRS distribution across the five clinical subtypes (Kruskal-Wallis $H = 15.29$, $p = 0.0041$), suggesting that the taxonomy captures part of the underlying genetic susceptibility. Additional analyses of *LRRK2* and *GBA* carrier status via chi-square tests further support differential genetic enrichment patterns, although limited allele frequencies constrain statistical power.

Fairness and demographic robustness are audited using a fairness-oriented index (FAIS) applied to the distribution of baseline subtypes across age, sex and site. Subtype assignments appear well balanced with respect to age, and sex-related asymmetries (e.g. higher representation of the severe subtype in women, greater representation of a motor-dominant subtype in men) are consistent with known epidemiological patterns rather than artefacts of the pipeline, which already controls for sex in the GAMLSS normalisation. A permutation test of ARI_{xc} (1,000 permutations) yields a non-significant p -value (0.334), indicating that, despite a positive and moderate inter-cohort agreement (Cohen’s $\kappa \approx 0.41$), the current OPDC sample ($N = 146$) is underpowered to demonstrate statistical generalisability at conventional thresholds; power calculations suggest that 400-600 OPDC patients would be required to achieve 80% power at the observed effect size.

Overall, this baseline clustering and validation pipeline delivers a statistically robust and clinically interpretable stratification of Parkinson’s disease into five subtypes with distinct initial profiles, subsequent progression trajectories, and convergent biological signatures across imaging and genetics. By coupling rigorous pre-processing (GAMLSS, MICE, ComBat-EB), consensus GMM clustering at baseline, SuStaIn-based progression modelling, and multi-level orthogonal validation, it addresses key methodological gaps in previous subtype studies, particularly in cross-cohort harmonisation and longitudinal validation. The resulting taxonomy offers a concrete framework for risk stratification, prognostic enrichment and future treatment targeting in early Parkinson’s disease, while also delineating clear requirements for external replication in larger, independent cohorts.